READING GROUP: TOPOLOGICAL CYCLIC HOMOLOGY
WINTER TERM 2026/2027

NIKLAS ARPPE AND LEWIS NGO

Algebraic K-theory is one of the fundamental invariants in modern mathematics. Since its introduction by Grothendieck, it has found applications throughout topology, algebraic geometry, and number theory. Unfortunately, it is also notoriously difficult to compute, and explicit calculations are often highly nontrivial even for relatively simple rings. We do not even know the algebraic K-theory of the integers yet!

The most successful approach to computing algebraic K-theory is through trace methods. Bökstedt's topological Hochschild homology THH and the subsequent theory of topological cyclic homology TC provide computable approximations to algebraic K-theory via the Dennis and cyclotomic trace maps. In particular, the trace map $K \longrightarrow \mathrm{TC}$ is a generalization of the classical Chern character. Combined with the Dundas–Goodwillie–McCarthy theorem, these have led to many computations of algebraic K-groups that were previously out of reach. More recently, trace methods have become an indispensable tool in arithmetic geometry through their connections with prismatic cohomology, syntomic cohomology, and p-adic Hodge theory in general.

The aim of this reading group is to develop these ideas carefully from the ground up. We will begin by introducing topological Hochschild homology THH, cyclotomic spectra, and the associated theories TC^-, TP, and TC. Along the way we will work through a number of fundamental computations, including Bökstedt periodicity, the calculation of the topological cyclic homology of $\mathbf{F}_p$ and perfect $\mathbf{F}_p$-algebras, and examples arising from Thom spectra. Besides illustrating the general theory, these examples introduce many of the computational techniques that make trace methods so effective, such as homotopy fixed point and Tate spectral sequences.

In the second half of the semester, we will move towards some of the most recent developments in the subject. We will study the Bhatt–Morrow–Scholze motivic filtrations on TC^- and TP, explain how they recover familiar objects such as the de Rham–Witt complex and syntomic cohomology, and discuss their role as the trace-theoretic analogue of the motivic filtration on algebraic K-theory. Using these one obtains a corresponding filtration on THH in the obvious way. Time permitting, we will conclude with the theorem of Clausen–Mathew–Morrow identifying p-completed algebraic K-theory with p-completed TC for a particular class of rings, which represents the current state of the art in using trace methods to understand algebraic K-theory in arithmetic settings.

Prerequisites

We will assume a solid understanding of higher category theory and higher algebra. Familiarity with algebraic K-theory and prismatic/syntomic cohomology will be useful for the later talks, but is not strictly necessary to follow the seminar.

Schedule

The reading group will meet on Mondays 16-18 in SemR N0.007.

No.	Date	Topic	Speaker
0.	Oct 14th	Preliminary meeting	Lewis and Niklas
1.	Oct 19th	Topological Hochschild homology	Yuxin
2.	Oct 26th	Bökstedt periodicity	Jordi
3.	Nov 2nd	Cyclotomic spectra and topological cyclic homology	Niklas
4.	Nov 16th	Topological cyclic homology of perfect $\mathbf{F}_p$-algebras	Philipp
5.	Nov 23rd	Thom spectra and group rings	Vieri
6.	Nov 30th	Algebraic K-theory and trace methods	Kevin
7.	Dec 7th	Introduction to prismatic cohomology	Vishal
8.	Dec 14th	More on prismatic cohomology	Niklas
9.	Jan 11th	The motivic filtration on TC^- and TP I	Lewis
10.	Jan 18th	The motivic filtration on TC^- and TP II	Daniel
11.	Jan 25th	Étale K-theory of Henselian pairs	Ostap

Syllabus

Talk 0: Preliminary meeting. We will provide an overview of the topic and assign talks. If you are interested in giving a talk, please attend the meeting. Any remaining talks will be available afterward, and you may contact the organizers directly if you would like to present.

Talk 1: Topological Hochschild homology. For an $\mathbf{E}_1$-ring R, define the *topological Hochschild homology spectrum* $\mathrm{THH}(R)$ as the geometric realization

$$\mathrm{THH}(R) := |\cdots \rightrightarrows R \otimes R \rightrightarrows R|.$$

Prove that this can equivalently be defined as $\mathrm{THH}(R) = R \otimes_{R \otimes R^{\mathrm{op}}} R$ (cf. [Zhu26, Lemma 4.1.12]). In particular, if R is an $\mathbf{E}_\infty$-ring, we can get rid of the op. Prove the McClure–Schwänzl–Vogt theorem [NS18, Proposition IV.2.2], identifying $\mathrm{THH}(R)$ as the free $\mathbf{E}_\infty$-ring with an S^1-action; this shows that $\mathrm{THH}(R) \simeq R^{\otimes \mathbf{T}}$. Show that we can define a natural relative version of THH. Show that THH is symmetric monoidal (cf. [Zhu26, Proposition 4.3.1]) and conclude by stating the THH of some common rings, e.g., $\mathbf{S}$, $\mathbf{Z}$, $\mathbf{Z}/p^n$.

Talk 2: Bökstedt periodicity. The goal of this talk will be to show that $\mathrm{THH}_*(\mathbf{F}_p) \cong \mathbf{F}_p[x]$ with $|x| = 2$ following [Zhu26, Section 4]. Prove that we can equivalently show that

$$\mathrm{THH}(\mathbf{F}_p) \simeq \mathrm{H}\mathbf{F}_p \otimes \Sigma^\infty_+ \Omega S^3$$

as $\mathrm{H}\mathbf{F}_p$-$\mathbf{E}_1$-algebras [Zhu26, Theorem 4.2.1]. By the definition of $\mathrm{THH}(\mathbf{F}_p)$, we are reduced to investigating $\mathrm{H}\mathbf{F}_p \otimes \mathrm{H}\mathbf{F}_p$. In particular, we will investigate $\pi_*(\mathrm{H}\mathbf{F}_p \otimes \mathrm{H}\mathbf{F}_p)$, which is the *dual Steenrod algebra*. Use the rest of the talk to introduce Dyer–Lashof operations and prove that we have an equivalence

$$\mathrm{H}\mathbf{F}_p \otimes \mathrm{H}\mathbf{F}_p \simeq \mathrm{H}\mathbf{F}_p \otimes \Sigma^\infty_+ \Omega^2 S^3$$

as $\mathrm{H}\mathbf{F}_p$-$\mathbf{E}_2$-algebras. Also mention the slightly different argument using the Hopkins–Mahowald theorem [KN, Theorem A.1] (this is outlined in [Zhu26, Remark 4.2.9]).

Talk 3: Cyclotomic spectra and topological cyclic homology. This talk equips THH with a cyclotomic structure. Begin by quickly reminding the audience of some of Borel equivariant homotopy theory and the Tate construction as in [Zhu26, Appendix A]. There is no need to discuss any genuine equivariant homotopy theory or other higher categorical coherence issues concerning the Tate construction. Introduce the ∞-category of cyclotomic spectra and define the topological cyclic homology spectrum $\mathrm{TC}(R)$ for an $\mathbf{E}_1$-ring R [NS18, Section II.1]. You should also introduce other important candidates of cyclotomic spectra such as TC^- and TP [oObe18, Talk 4].

Talk 4: Topological cyclic homology of perfect $\mathbf{F}_p$-algebras. Compute the topological cyclic homology of $\mathbf{F}_p$ following [Zhu26, Section 6.1]. To do this, exhibit $\mathrm{TC}(\mathbf{F}_p)$ as the equalizer

$$\mathrm{TC}(\mathbf{F}_p) \simeq \mathrm{eq}(\mathrm{TC}^-(\mathbf{F}_p) \rightrightarrows \mathrm{TP}(\mathbf{F}_p)^\wedge_p);$$

hence, it will suffice to understand $\mathrm{TC}^-_*(\mathbf{F}_p)$, $\mathrm{TP}_*(\mathbf{F}_p)$, and the maps relating them. For the first two, recall [Zhu26, Theorem 5.2.9] and [Zhu26, Theorem 5.3.12]. The canonical map can is clear, so spend the remaining time on understanding $\varphi\colon \mathrm{TC}^-_*(\mathbf{F}_p) \longrightarrow \mathrm{TP}_*(\mathbf{F}_p)$ and conclude the main proof. Remark how this relates to the p-completed algebraic K-theory $K_*(\mathbf{F}_p)^\wedge_p$.

Generalize the preceding computation to perfect $\mathbf{F}_p$-algebras following [Zhu26, Section 6.2]. Recall Bökstedt's theorem for perfect $\mathbf{F}_p$-algebras, i.e., that $\mathrm{THH}_*(k) \cong k[x]$ for k a perfect $\mathbf{F}_p$-algebra. Again, proceed by first computing $\mathrm{TC}^-_*(k)$ and $\mathrm{TP}_*(k)$, and then combine these to deduce $\mathrm{TC}_*(k)$. For this, recall the definition of the ring of Witt vectors $W(k)$ and the *homotopy fixed point spectral sequence*. For the latter, see [Zhu26, Theorem 5.2.7]. While you are at it, define the Tate spectral sequence for future reference [Zhu26, Proposition 5.3.3].

Talk 5: Thom spectra and group rings. The purpose of this talk is to use Thom spectra as a source of explicit computations of THH. Begin by recalling the Thom spectrum $\mathrm{M}f$ associated with a map $f\colon X \longrightarrow \mathrm{Pic}(R)$, and recover some basic examples of Thom spectra [ABG+14, Definition 1.4]. The given reference writes R-line, but you should write $\mathrm{Pic}(R)$. Formulate general results about $\mathrm{THH}(\mathrm{M}f/R)$ and use this to state the general formula for THH of a Thom spectrum. In particular, apply this to compute $\mathrm{THH}_*(\mathrm{MU})$ [KN, Section 4]. For a discrete group G, view the spherical group ring $\mathbf{S}[G] \simeq \Sigma^\infty_+ G$ as the Thom spectrum of the trivial map $G \longrightarrow \mathrm{Pic}(\mathbf{S})$, and use the Thom spectrum formula to deduce $\mathrm{THH}(\mathbf{S}[G]) \simeq \Sigma^\infty_+ LBG$, where L denotes the free loop space functor and the circle action rotates loops. Finish the talk by working out the case $G = \mathbf{Z}$ explicitly and concluding with the analogous computation for $\mathbf{S}[\mathbf{N}]$ using the cyclic bar construction. In particular, emphasize the geometric content of the free loop space description: the decomposition of $LB\mathbf{Z} \simeq LS^1$ by winding number makes both the components of $\mathrm{THH}(\mathbf{S}[\mathbf{Z}])$ and its S^1-action explicit. [oTal24, Talk 1 – Catherine Li] may also be a helpful reference. This talk may require some familiarity with the fundamental Thom spectrum MU, though detailed knowledge is not necessary.

Talk 6: Algebraic K-theory and trace methods. Possibly the biggest reason for studying THH and its variants is the relationship with algebraic K-theory. The purpose of this talk is to shed some light on this connection. A good reference for this talk is Gepner's lecture in [oObe18]; see also [oTal24, Lecture 15]. Briefly recall Waldhausen's S-construction of algebraic K-theory (cf. [HMC24, Section 5.2] or [HW21, Construction IV.5]). Define the *Dennis trace* as the unique lax symmetric monoidal map $K \longrightarrow \mathrm{THH}$ corresponding to $1 \in \pi_0\mathbf{S} \cong \mathbf{Z}$ under the correspondence in the Blumberg-Gepner-Tabuada theorem. Since TC is not an additive invariant, we will have to work harder to construct the *cyclotomic trace* $K \longrightarrow \mathrm{TC}$. Do this by constructing the invariants

$$\mathrm{TC}^n := \mathrm{eq}(\mathrm{TR}^n \rightrightarrows \mathrm{TR}^{n-1}),$$

and applying the Blumberg-Gepner-Tabuada theorem to obtain a compatible system of maps $K \longrightarrow \mathrm{TC}^n$. Conclude by stating the Dundas–Goodwillie–McCarthy theorem.

Talk 7: Introduction to prismatic cohomology. The purpose of this talk is to give a quick introduction to prismatic cohomology. For this, you will probably want to follow [BS22] or Kedlaya's notes. Start by defining the ring of Witt vectors associated with a perfect $\mathbf{F}_p$-algebra and provide many examples. Define δ-rings and use these to construct prisms. Define the prismatic site and state the usual comparison theorems with étale, crystalline, and de Rham cohomology. Time permitting, define the quasi-syntomic site, for example, following [BL22, Appendix C].

Talk 8: More on prismatic cohomology. Define the Nygaard filtration on prismatic cohomology (cf. [BS22, Section 12, Section 15]), and use this to define syntomic cohomology, at least in the

case when the base is a perfectoid ring (see the discussion at the beginning of [BS22, Section 14]). Coordinate with the speaker of Talk 10 regarding any additional necessary content.

Talk 9: The motivic filtrations on TC^- and TP I. In a nutshell, the goal of this talk will be to discuss [BMS19, Theorem 1.12]. Following [BMS19, Section 7], construct the motivic filtrations on TC^- and TP, which are meant to be analogs of the motivic filtration on algebraic K-theory. You will certainly not be able to prove everything, so focus on communicating the key results. Another good reference for this talk is Le Bras' lecture in [oObe18].

Talk 10: The motivic filtrations on TC^- and TP II. Having constructed the motivic filtrations in the previous talk, we will now investigate the characteristic p case. Following [BMS19, Secton 8], introduce the Nygaard filtration on the de Rham–Witt complex and its derived variant, and state how this relates to the Nygaard filtration on primatic cohomology [BMS19, Theorem 8.17]. State [BMS19, Corollary 8.18], and use the remaining time to discuss the relation to syntomic cohomology [BMS19, Section 8.4]. Again, you will certainly need to omit many proofs, so focus on communicating the relevant content.

Talk 11: Étale K-theory of Henselian pairs. Besides the Dundas–Goodwillie–McCarthy theorem, one of the strongest results we have available for computing K-theory is due to Clausen–Mathew–Morrow: Let R be a strictly henselian local ring with residue field of characteristic $p > 0$. Then, the cyclotomic trace $K(R) \longrightarrow \mathrm{TC}(R)$ is a p-adic equivalence [CMM20, Theorem 6.1]. The purpose of this talk is to sketch the proof of this result. The organization of this talk is up to the speaker; you need to decide how much background to recall and how much detail to include.

References

[ABG+14] Matthew Ando et al., *An ∞-categorical approach to R-line bundles, R-module Thom spectra, and twisted R-homology*, 2014. arXiv: 1403.4325 [math.AT].

[BL22] Bhargav Bhatt and Jacob Lurie, *Absolute prismatic cohomology*, 2022. arXiv: 2201.06120 [math.AG].

[BMS19] Bhargav Bhatt, Matthew Morrow, and Peter Scholze, *Topological Hochschild homology and integral p-adic Hodge theory*, 2019. arXiv: 1802.03261 [math.AG].

[BS22] Bhargav Bhatt and Peter Scholze, *Prisms and Prismatic Cohomology*, 2022. arXiv: 1905.08229 [math.AG].

[CMM20] Dustin Clausen, Akhil Mathew, and Matthew Morrow, *K-theory and topological cyclic homology of henselian pairs*, 2020. arXiv: 1803.10897 [math.KT].

[HMC24] Kaif Hilman and Jonas McCandless, *Introduction to algebraic K-theory*, 2024.

[HW21] Fabian Hebestreit and Ferdinand Wagner, *Algebraic and Hermitian K-theory*, 2021.

[KN] Achim Krause and Thomas Nikolaus, *Lectures on topological Hochschild homology and cyclotomic spectra*, Unpublished lecture notes.

[NS18] Thomas Nikolaus and Peter Scholze, *On topological cyclic homology*, Acta Mathematica, vol. 221, no. 2, pp. 203–409, 2018.

[oObe18] Participants of Oberwolfach Arbeitsgemeinschaft 2018, *Oberwolfach arbeitsgemeinschaft report: Topological cyclic topology*, Workshop notes, 2018.

[oTal24] Participants of Talbot 2024, *Talbot 2024: Topological cyclic homology of ring spectra*, Unofficial notes, 2024.

[Zhu26] Qi Zhu, *Venturing into Topological Cyclic Homology*, Unofficial notes, 2026.